Computer Graphics

Section: H

Title: Village scenario.

Background: This is a project where we've demonstrated a village scenario. In our project, there are some house, trees, hills, boats in river, clouds and birds are moving in the sky. We have used "CodeBlocks" as platform and C++ as programming language.

Features: Our project is about village scenery. Here we have shown sunset and a moonlit night. This scenery consists of 2 huts, 1 tree, 3 boats floating on a river, some hills, moving clouds and flying birds in the sky. There is a cow in the field just in front the house. Here we have shown a boat which can be moved in two direction. We have to press "R" to move the boat in right direction and "L" to move in left. The night view will automatically appear after sunset and then the moon will be rising.

Individual contribution:

Features	Members
Sun, moon drawing and movement with night view and river	Md. Ahasanul Islam (19-40396-1)
Tree, House-1 and road with ground	Mehedi Hasan (18-36328-1)
River with moving boat and hills drawing	Israt Jahan Pritha (18-39023-3)
Birds and clouds drawing and movement, fence drawing	Zareen Zia (17-35760-3)
House-2, cow and boat	Arnab Ibna Hasan (17-35690-3)

Features Pic:

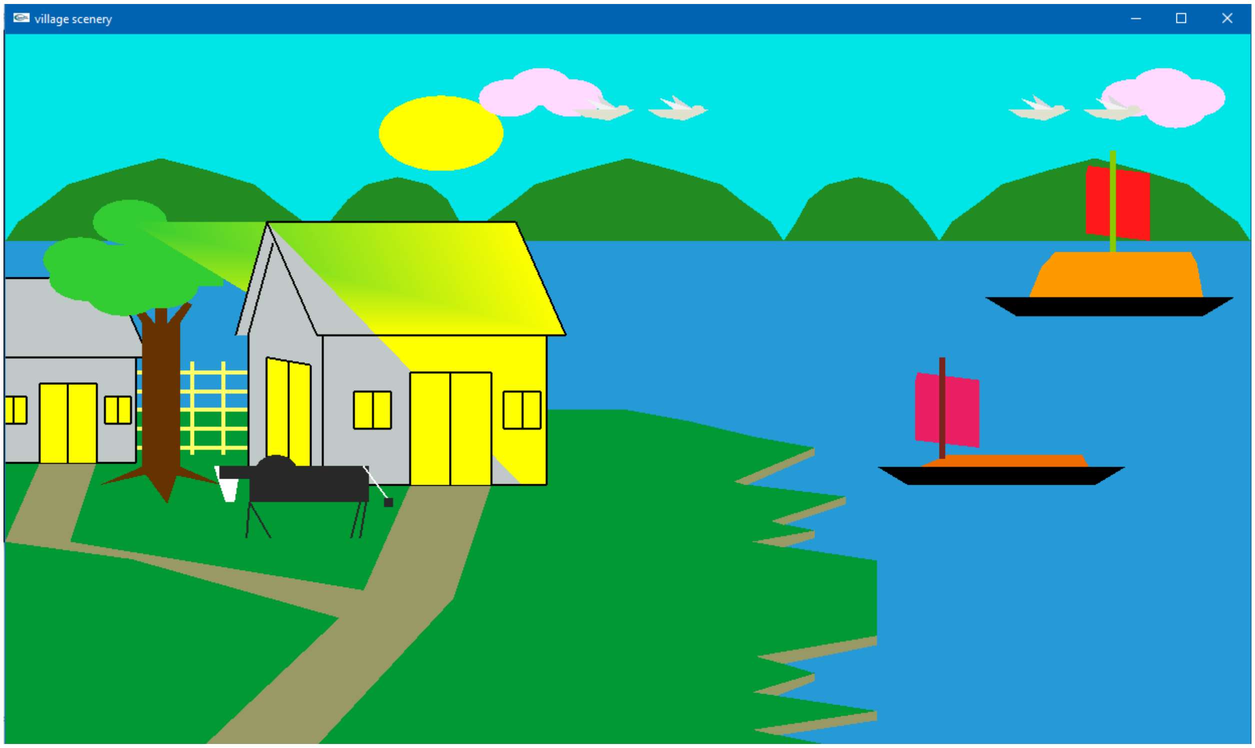

Fig1: Day view of village scenario

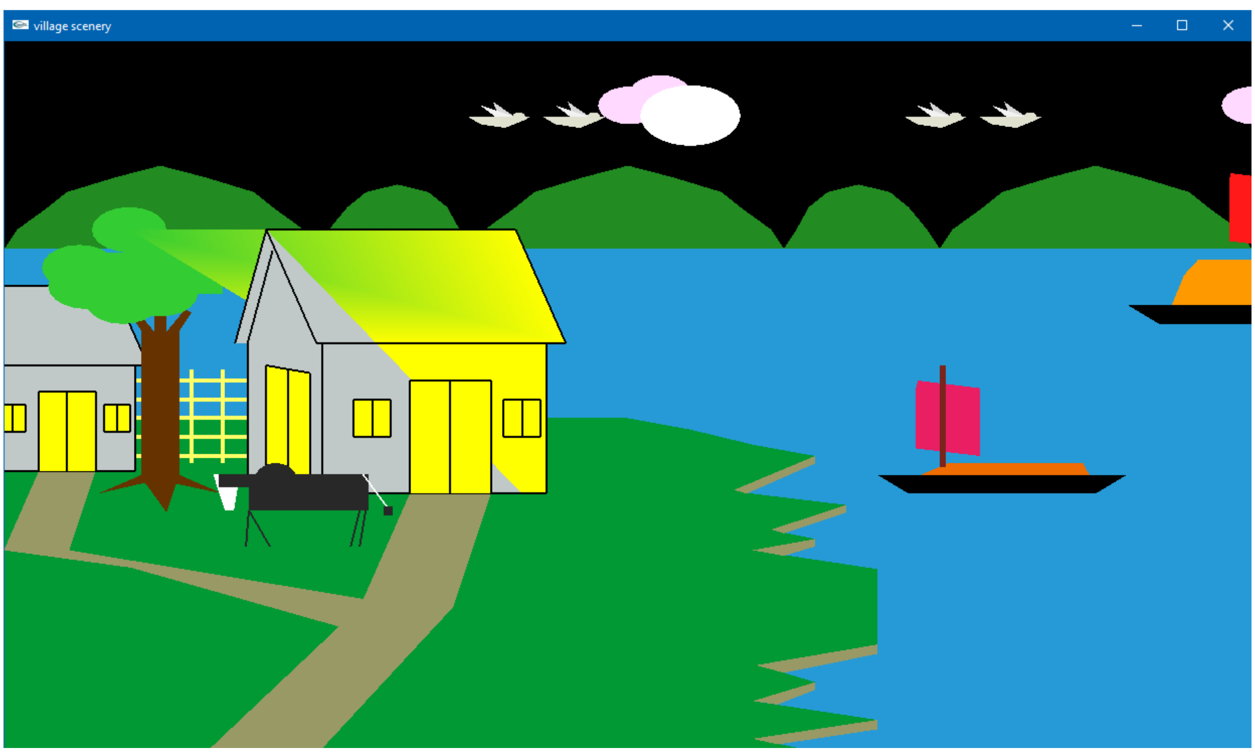

Fig2: Night view of village scenario

Features Code:

```
#include<cstdio>

#include <windows.h>

#include<math.h>

#include <vector>

#include <cstdlib>

# define PI 3.14159265358979323846

#include <GL/gl.h>

#include <GL/glut.h>

#include<MMSystem.h>


float speedb = 0.002f;

float moveb = 0.00f;

float h=0.0,i=0.9,j=0.9;


GLfloat position22 = 0.0f;

GLfloat speed22 = 0.007f;

void birdd(int value) {

    if(position22 > 1.0)

        position22 =-1.0f;

    position22 += speed22;

                glutPostRedisplay();

                glutTimerFunc(100, birdd, 0);

}

GLfloat position4 = 0.0f;

GLfloat speed4 =-0.01f;


GLfloat position2 = 0.0f;

GLfloat speed2 = 0.004f;

void cloud(int value) {
```

```
    if(position2 > 1.0)
        position2 =-1.0f;
    position2 += speed2;
            glutPostRedisplay();
            glutTimerFunc(100, cloud, 0);
}

GLfloat sunPosition = 0.0f;
GLfloat sunSpeed = 0.002f;

void updateSun(int value)
{
    if(sunPosition < -.9)
        {
            sunPosition = -.9;


        }

    sunPosition -= sunSpeed;

    glutPostRedisplay();

    glutTimerFunc(100, updateSun, 0);
}

GLfloat moonPosition = 0.0f;
GLfloat moonSpeed = 0.003f;

void updateMoon(int value)
{
    moonPosition += moonSpeed;
```

```
    if(moonPosition > 0.3)
        {
            moonPosition = 0.3;
        }
        moonPosition += moonSpeed;

    glutPostRedisplay();

            glutTimerFunc(100, updateMoon, 0);
}

void cloud1()
{

    int i;

            GLfloat x=.5f; GLfloat y=.86f; GLfloat radius =.05f;
            int triangleAmount = 20;
            GLfloat twicePi = 2.0f * PI;

            glBegin(GL_TRIANGLE_FAN);
                glColor3ub(255, 217, 255);
                        glVertex2f(x, y); // center of circle
                        for(i = 0; i <= triangleAmount;i++) {
                                glVertex2f(
                            x + (radius * cos(i *  twicePi / triangleAmount)),
                                y + (radius * sin(i * twicePi / triangleAmount))
                                );
                        }
            glEnd();
```

```
GLfloat a=.55f; GLfloat b=.83f;

        glBegin(GL_TRIANGLE_FAN);
          glColor3ub(255, 217, 255);
                glVertex2f(a, b); // center of circle
                for(i = 0; i <= triangleAmount;i++) {
                        glVertex2f(
                      a + (radius * cos(i *  twicePi / triangleAmount)),
                          b + (radius * sin(i * twicePi / triangleAmount))
                        );
                }
        glEnd();

        GLfloat c=.45f; GLfloat d=.83f;

        glBegin(GL_TRIANGLE_FAN);
          glColor3ub(255, 217, 255);
                glVertex2f(c, d); // center of circle
                for(i = 0; i <= triangleAmount;i++) {
                        glVertex2f(
                      c + (radius * cos(i *  twicePi / triangleAmount)),
                          d + (radius * sin(i * twicePi / triangleAmount))
                        );
                }
        glEnd();

        GLfloat e=.52f; GLfloat f=.8f;

        glBegin(GL_TRIANGLE_FAN);
          glColor3ub(255, 217, 255);
                glVertex2f(e, f); // center of circle
```

```
                for(i = 0; i <= triangleAmount;i++) {
                    glVertex2f(
                   e + (radius * cos(i *  twicePi / triangleAmount)),
                     f+ (radius * sin(i * twicePi / triangleAmount))
                    );
                }
        glEnd();




}


void cloud2()
{


   int i;


        GLfloat x=-.5f; GLfloat y=.86f; GLfloat radius =.05f;
        int triangleAmount = 20;
        GLfloat twicePi = 2.0f * PI;


        glBegin(GL_TRIANGLE_FAN);
           glColor3ub(255, 217, 255);
                glVertex2f(x, y); // center of circle
                for(i = 0; i <= triangleAmount;i++) {
                    glVertex2f(
                   x + (radius * cos(i *  twicePi / triangleAmount)),
                     y + (radius * sin(i * twicePi / triangleAmount))
                    );
                }
        glEnd();
```

```
GLfloat a=-.55f; GLfloat b=.83f;

    glBegin(GL_TRIANGLE_FAN);
      glColor3ub(255, 217, 255);
        glVertex2f(a, b); // center of circle
        for(i = 0; i <= triangleAmount;i++) {
            glVertex2f(
          a + (radius * cos(i *  twicePi / triangleAmount)),
              b + (radius * sin(i * twicePi / triangleAmount))
            );
        }
    glEnd();

    GLfloat c=-.45f; GLfloat d=.83f;

    glBegin(GL_TRIANGLE_FAN);
      glColor3ub(255, 217, 255);
        glVertex2f(c, d); // center of circle
        for(i = 0; i <= triangleAmount;i++) {
            glVertex2f(
          c + (radius * cos(i *  twicePi / triangleAmount)),
              d + (radius * sin(i * twicePi / triangleAmount))
            );
        }
    glEnd();



}
void cloud3()
{
```

```
int i;

        GLfloat x=0.0f; GLfloat y=.86f; GLfloat radius =.05f;
        int triangleAmount = 20;
        GLfloat twicePi = 2.0f * PI;

        glBegin(GL_TRIANGLE_FAN);
          glColor3ub(255, 217, 255);
                glVertex2f(x, y); // center of circle
                for(i = 0; i <= triangleAmount;i++) {
                        glVertex2f(
                      x + (radius * cos(i *  twicePi / triangleAmount)),
                          y + (radius * sin(i * twicePi / triangleAmount))
                        );
                }
        glEnd();



        GLfloat e=.02f; GLfloat f=.8f;

        glBegin(GL_TRIANGLE_FAN);
          glColor3ub(255, 217, 255);
                glVertex2f(e, f); // center of circle
                for(i = 0; i <= triangleAmount;i++) {
                        glVertex2f(
                      e + (radius * cos(i *  twicePi / triangleAmount)),
                          f+ (radius * sin(i * twicePi / triangleAmount))
                        );
                }
```

```
            glEnd();

            GLfloat g=.1f; GLfloat h=.82f;

            glBegin(GL_TRIANGLE_FAN);
               glColor3ub(255, 217, 255);
                        glVertex2f(g, h); // center of circle
                        for(i = 0; i <= triangleAmount;i++) {
                                    glVertex2f(
                                  g + (radius * cos(i *  twicePi / triangleAmount)),
                                       h+ (radius * sin(i * twicePi / triangleAmount))
                                    );
                        }
            glEnd();


}
void sky()
{
   glColor3f(h,i,j);
   glBegin(GL_QUADS);
            glVertex2f(-1.0f, 0.45f);
            glVertex2f(1.0f, 0.45f);
   glVertex2f(1.0f, 1.0f);
            glVertex2f(-1.0f, 1.0f);
   glEnd();
}

void hills()
{
```

```
glBegin(GL_POLYGON);
glColor3ub(34,139,34);
glVertex2f(-1.0f,0.45f);
glVertex2f(-0.98f, 0.5f);
glVertex2f(-0.93f, 0.56);
glVertex2f(-0.9f,0.6f);
glVertex2f(-0.82f, 0.64);
glVertex2f(-0.75f, 0.67);
glVertex2f(-0.68f, 0.64);
glVertex2f(-0.6f,0.6f);
glVertex2f(-0.57f, 0.56);
glVertex2f(-0.52f, 0.5f);
glVertex2f(-0.5f,0.45f);
glVertex2f(-1.0f,0.45f);
glEnd();

glBegin(GL_POLYGON);
glColor3ub(34,139,34);
glVertex2f(1.0f,0.45f);
glVertex2f(0.98f, 0.5f);
glVertex2f(0.93f, 0.56);
glVertex2f(0.9f,0.6f);
glVertex2f(0.82f, 0.64);
glVertex2f(0.75f, 0.67);
glVertex2f(0.68f, 0.64);
glVertex2f(0.6f,0.6f);
glVertex2f(0.57f, 0.56);
glVertex2f(0.52f, 0.5f);
glVertex2f(0.5f,0.45f);
glVertex2f(1.0f,0.45f);
glEnd();
```

```
            glBegin(GL_POLYGON);
            glColor3ub(34,139,34);
            glVertex2f(-0.5f,0.45f);
            glVertex2f(-0.48f, 0.5f);
glVertex2f(-0.45f, 0.56);
glVertex2f(-0.42f,0.6f);
glVertex2f(-0.37f, 0.62);
glVertex2f(-0.32f, 0.6);
glVertex2f(-0.29f, 0.56f);
 glVertex2f(-0.27f, 0.5f);
glVertex2f(-0.25f,0.45f);
glEnd();

glBegin(GL_POLYGON);
            glColor3ub(34,139,34);
            glVertex2f(0.5f,0.45f);
            glVertex2f(0.48f, 0.5f);
glVertex2f(0.45f, 0.56);
glVertex2f(0.42f,0.6f);
glVertex2f(0.37f, 0.62);
glVertex2f(0.32f, 0.6);
glVertex2f(0.29f, 0.56f);
 glVertex2f(0.27f, 0.5f);
glVertex2f(0.25f,0.45f);
glEnd();

glBegin(GL_POLYGON);
            glColor3ub(34,139,34);
            glVertex2f(-.25f,0.45f);
            glVertex2f(-0.23f, 0.5f);
```

```
   glVertex2f(-0.18f, 0.56);
   glVertex2f(-0.15f,0.6f);
   glVertex2f(-0.07f, 0.64);
   glVertex2f(-0.00f, 0.67);
   glVertex2f(0.07f, 0.64);
   glVertex2f(0.15f,0.6f);
   glVertex2f(0.18f, 0.56);
   glVertex2f(0.23f, 0.5f);
   glVertex2f(.25f,0.45f);
            glEnd();

}

void bird()
{
      int i;

            GLfloat mm=0.182f; GLfloat nn=.801f; GLfloat radiusmm =.01f;
            int triangleAmount = 20;
            GLfloat twicePi = 2.0f * PI;

            glBegin(GL_TRIANGLE_FAN);
               glColor3ub(225, 225, 208);
                        glVertex2f(mm, nn); // center of circle
                        for(i = 0; i <= triangleAmount;i++) {
                                    glVertex2f(
                                 mm + (radiusmm * cos(i *  twicePi / triangleAmount)),
                                      nn + (radiusmm * sin(i * twicePi / triangleAmount))
                                    );
                        }
            glEnd();
```

```
glBegin(GL_POLYGON);
glColor3ub(225, 225, 208 );
glVertex2f(0.1f,0.8f);
glVertex2f(0.11f,0.79f);
glVertex2f(0.12f,0.78f);
glVertex2f(0.16f,0.77f);
glVertex2f(0.19f,0.79f);
glVertex2f(0.201f,0.8f);
glEnd();

glBegin(GL_TRIANGLES);
glColor3ub(217, 217, 217);
glVertex2f(0.175f,0.8f);
glVertex2f(0.15f,0.8f);
glVertex2f(0.14f,0.84f);
glEnd();


glBegin(GL_TRIANGLES);
glColor3ub(242, 242, 242 );
glVertex2f(0.175f,0.8f);
glVertex2f(0.144f,0.8f);
glVertex2f(0.12f,0.83f);
glEnd();
        /////2nd bird////
        glBegin(GL_POLYGON);
glColor3ub(225, 225, 208 );
glVertex2f(-0.02f,0.8f);
glVertex2f(-0.01f,0.79f);
glVertex2f(0.0f,0.78f);
glVertex2f(0.04f,0.77f);
```

```
glVertex2f(0.07f,0.79f);
glVertex2f(0.081f,0.8f);
glEnd();

glBegin(GL_TRIANGLES);
glColor3ub(217, 217, 217);
glVertex2f(0.055f,0.8f);
glVertex2f(0.03f,0.8f);
glVertex2f(0.02f,0.84f);
glEnd();



glBegin(GL_TRIANGLES);
glColor3ub(242, 242, 242 );
glVertex2f(0.055f,0.8f);
glVertex2f(0.024f,0.8f);
glVertex2f(0.0f,0.83f);
glEnd();

    GLfloat mmm=0.062f; GLfloat nnn=.801f; GLfloat radiusmmm =.01f;

    glBegin(GL_TRIANGLE_FAN);
      glColor3ub(225, 225, 208);
            glVertex2f(mmm, nnn); // center of circle
            for(i = 0; i <= triangleAmount;i++) {
                    glVertex2f(
                  mmm + (radiusmmm * cos(i *  twicePi / triangleAmount)),
                      nnn + (radiusmmm * sin(i * twicePi / triangleAmount))
                    );
            }
    glEnd();
```

```
/////3rd bird/////
glBegin(GL_POLYGON);
glColor3ub(225, 225, 208 );
glVertex2f(-0.72f,0.8f);
glVertex2f(-0.71f,0.79f);
glVertex2f(-0.7f,0.78f);
glVertex2f(-0.66f,0.77f);
glVertex2f(-0.63f,0.79f);
glVertex2f(-0.619f,0.8f);
glEnd();

glBegin(GL_TRIANGLES);
glColor3ub(217, 217, 217);
glVertex2f(-0.645f,0.8f);
glVertex2f(-0.67f,0.8f);
glVertex2f(-0.68f,0.84f);
glEnd();

glBegin(GL_TRIANGLES);
glColor3ub(242, 242, 242 );
glVertex2f(-0.645f,0.8f);
glVertex2f(-0.676f,0.8f);
glVertex2f(-0.7f,0.83f);
glEnd();

GLfloat mmmm=-0.638f; GLfloat nnnn=.801f;

glBegin(GL_TRIANGLE_FAN);
   glColor3ub(225, 225, 208);
      glVertex2f(mmmm,nnnn); // center of circle
      for(i = 0; i <= triangleAmount;i++) {
```

```
                        glVertex2f(
                      mmmm + (radiusmmm * cos(i *  twicePi / triangleAmount)),
                          nnnn + (radiusmmm * sin(i * twicePi / triangleAmount))
                        );
                }
        glEnd();
        ////4th bird////
        GLfloat mmmmm=-0.518f; GLfloat nnnnn=.801f;

        glBegin(GL_TRIANGLE_FAN);
          glColor3ub(225, 225, 208);
                glVertex2f(mmmmm, nnnnn); // center of circle
                for(i = 0; i <= triangleAmount;i++) {
                        glVertex2f(
                      mmmmm + (radiusmm * cos(i *  twicePi / triangleAmount)),
                          nnnnn + (radiusmm * sin(i * twicePi / triangleAmount))
                        );
                }
        glEnd();
glBegin(GL_POLYGON);
glColor3ub(225, 225, 208 );
glVertex2f(-0.6f,0.8f);
glVertex2f(-0.59f,0.79f);
glVertex2f(-0.58f,0.78f);
glVertex2f(-0.54f,0.77f);
glVertex2f(-0.51f,0.79f);
glVertex2f(-0.499f,0.8f);
glEnd();

glBegin(GL_TRIANGLES);
glColor3ub(217, 217, 217);
```

```
    glVertex2f(-0.525f,0.8f);
    glVertex2f(-0.55f,0.8f);
    glVertex2f(-0.56f,0.84f);
    glEnd();



    glBegin(GL_TRIANGLES);
    glColor3ub(242, 242, 242 );
    glVertex2f(-0.525f,0.8f);
    glVertex2f(-0.556f,0.8f);
    glVertex2f(-0.58f,0.83f);
    glEnd();



}



void sun()
{
    glPushMatrix();
    glTranslatef(0.5f,sunPosition, 0.0f);

    float x1,y1,x2,y2;
    float a;
    double radius=0.1;

    x1=-0.8, y1= 0.9;

    glBegin(GL_TRIANGLE_FAN);
    glColor3f(239,255,0);
    for (a = 0.0f ; a < 360.0f ; a+=0.2)
```

```
        {
            x2 = x1+sin(a)*radius;
            y2 = y1+cos(a)*radius;
            glVertex2f(x2,y2);
        }

      glEnd();

   glPopMatrix();
}



void moon()
{
   glPushMatrix();
   glTranslatef(0.9f,moonPosition, 0.0f);
   float x1,y1,x2,y2;
   float a;
   double radius=0.08;

   x1=-0.8, y1= 0.5;

   glBegin(GL_TRIANGLE_FAN);
   glColor3f(255,255,255);
   for (a=0.0f; a<360.0f; a+=0.2)
      {
         x2 = x1+sin(a)*radius;
         y2 = y1+cos(a)*radius;
         glVertex2f(x2,y2);
      }
```

```
        glEnd();

    glPopMatrix();
}


void ground()
{
    glBegin(GL_POLYGON);
    glColor3ub(0, 153, 51);
                glVertex2f(-1.0f,-1.0f);
                glVertex2f(-1.0f,0.0f);
                glVertex2f(0.0f, 0.0f);
    glVertex2f(0.1f, -0.03);
    glVertex2f(0.2f, -0.07);
    glVertex2f(0.3f, -0.1);
    glVertex2f(0.2f, -0.13);
    glVertex2f(0.1f, -0.17);
    glVertex2f(0.2f, -0.2);
    glVertex2f(0.35f, -0.23);
    glVertex2f(0.25f, -0.25);
    glVertex2f(0.18f, -0.28);
    glVertex2f(0.3f, -0.32);
    glVertex2f(0.2f, -0.35);
    glVertex2f(0.4f, -0.4);
    glVertex2f(0.4f, -0.6);
    glVertex2f(0.2f, -0.65);
    glVertex2f(0.3f, -0.7);
    glVertex2f(0.2f, -0.75);
    glVertex2f(0.4f, -0.8);
    glVertex2f(0.2f, -0.85);
```

```
glVertex2f(0.35f, -0.9);
glVertex2f(0.25f, -0.95);
glVertex2f(0.4f, -1.0);
glEnd();

glBegin(GL_QUADS);
        glColor3ub(153, 153, 102);
glVertex2f(0.17f, -0.19);
glVertex2f(0.19f, -0.2f);
glVertex2f(0.3f, -0.12f);
glVertex2f(0.3f, -0.1);
glEnd();

glBegin(GL_QUADS);
        glColor3ub(153, 153, 102);
         glVertex2f(0.23f, -0.295);
         glVertex2f(0.25f, -0.305f);
         glVertex2f(0.35f, -0.25f);
        glVertex2f(0.35f, -0.23);
glEnd();

glBegin(GL_QUADS);
        glColor3ub(153, 153, 102);
        glVertex2f(0.3f, -0.32);
        glVertex2f(0.3f, -0.34);
glVertex2f(0.25f, -0.365f);
glVertex2f(0.2f, -0.35);
glEnd();

glBegin(GL_QUADS);
        glColor3ub(153, 153, 102);
```

```
            glVertex2f(0.205f, -0.655);
            glVertex2f(0.4f, -0.6);
            glVertex2f(0.4f, -0.625);
            glVertex2f(0.25f, -0.675);
    glEnd();

    glBegin(GL_QUADS);
            glColor3ub(153, 153, 102);
            glVertex2f(0.3f, -0.7);
            glVertex2f(0.3f, -0.72);
            glVertex2f(0.24f, -0.7595);
    glVertex2f(0.2f, -0.75);
    glEnd();

    glBegin(GL_QUADS);
            glColor3ub(153, 153, 102);
            glVertex2f(0.4f, -0.8);
            glVertex2f(0.4f, -0.825);
            glVertex2f(0.24f, -0.865);
    glVertex2f(0.2f, -0.85);
    glEnd();

    glBegin(GL_QUADS);
            glColor3ub(153, 153, 102);
            glVertex2f(0.35f, -0.9);
            glVertex2f(0.35f, -0.925);
            glVertex2f(0.27f, -0.958);
    glVertex2f(0.25f, -0.948);
    glEnd();
}
void river()
```

```
{
  glBegin(GL_QUADS);
  glColor3ub(38, 154, 214);
        glVertex2f(-1.0f,0.45f);
        glVertex2f(1.0f,0.45f);
        glVertex2f(1.0f,-1.0f);
   glVertex2f(-1.0f,-1.0f);
   glEnd();
}

void tree()
{
   glBegin(GL_POLYGON);
   glColor3ub(102, 51, 0);
   glVertex2f(-0.72f,-0.15f);
   glVertex2f(-0.65f,-0.2f);
   glVertex2f(-0.735f,-0.17f);
   glVertex2f(-0.74f,-0.25f);
   glVertex2f(-0.775f,-0.17f);
   glVertex2f(-0.85f,-0.2f);
   glVertex2f(-0.78f,-0.15f);

   glEnd();

   glBegin(GL_QUADS);
   glColor3ub(102, 51, 0);
   glVertex2f(-0.78f,-0.15f);
   glVertex2f(-0.78f,0.23f);
   glVertex2f(-0.72f,0.23f);
   glVertex2f(-0.72f,-0.15f);
   glEnd();
```

```
glBegin(GL_QUADS);
glColor3ub(102, 51, 0);
glVertex2f(-0.76f,0.23f);
glVertex2f(-0.76f,0.3f);
glVertex2f(-0.74f,0.3f);
glVertex2f(-0.74f,0.23f);
glEnd();
glBegin(GL_QUADS);
glColor3ub(102, 51, 0);
glVertex2f(-0.74f,0.23f);
glVertex2f(-0.71f,0.29f);
glVertex2f(-0.7f,0.28f);
glVertex2f(-0.72f,0.23f);
glEnd();
glBegin(GL_QUADS);
glColor3ub(102, 51, 0);
glVertex2f(-0.78f,0.23f);
glVertex2f(-0.8f,0.28f);
glVertex2f(-0.79f,0.29f);
glVertex2f(-0.76f,0.23f);
glEnd();

 int i;

        GLfloat x=-.75f; GLfloat y=.33f; GLfloat radius =.06f;
        int triangleAmount = 20;
        GLfloat twicePi = 2.0f * PI;

        glBegin(GL_TRIANGLE_FAN);
           glColor3ub(51, 204, 51);
                   glVertex2f(x, y);
```

```
            for(i = 0; i <= triangleAmount;i++) {
                    glVertex2f(
                  x + (radius * cos(i *  twicePi / triangleAmount)),
                      y + (radius * sin(i * twicePi / triangleAmount))
                    );
            }
glEnd();



GLfloat c=-.81f; GLfloat d=.31f;

glBegin(GL_TRIANGLE_FAN);
  glColor3ub(51, 204, 51);
            glVertex2f(c, d); // center of circle
            for(i = 0; i <= triangleAmount;i++) {
                    glVertex2f(
                  c + (radius * cos(i *  twicePi / triangleAmount)),
                      d + (radius * sin(i * twicePi / triangleAmount))
                    );
            }
glEnd();

GLfloat e=-.87f; GLfloat f=.35f;

glBegin(GL_TRIANGLE_FAN);
  glColor3ub(51, 204, 51);
            glVertex2f(e, f); // center of circle
            for(i = 0; i <= triangleAmount;i++) {
                    glVertex2f(
                  e + (radius * cos(i *  twicePi / triangleAmount)),
```

```
                    f+ (radius * sin(i * twicePi / triangleAmount))
                );
        }
glEnd();

GLfloat a1=-.61f; GLfloat b1=.4f;

glBegin(GL_TRIANGLE_FAN);
  glColor3ub(51, 204, 51);
        glVertex2f(a1, b1); // center of circle
        for(i = 0; i <= triangleAmount;i++) {
                glVertex2f(
              a1 + (radius * cos(i *  twicePi / triangleAmount)),
                    b1 + (radius * sin(i * twicePi / triangleAmount))
                );
        }
glEnd();

GLfloat c1=-.88f; GLfloat d1=.4f;

glBegin(GL_TRIANGLE_FAN);
  glColor3ub(51, 204, 51);
        glVertex2f(c, d); // center of circle
        for(i = 0; i <= triangleAmount;i++) {
                glVertex2f(
              c1 + (radius * cos(i *  twicePi / triangleAmount)),
                    d1 + (radius * sin(i * twicePi / triangleAmount))
                );
        }
glEnd();
```

```
        GLfloat e12=-.75f; GLfloat f12=.44f;

        glBegin(GL_TRIANGLE_FAN);
          glColor3ub(51, 204, 51);
                glVertex2f(e12, f12); // center of circle
                for(i = 0; i <= triangleAmount;i++) {
                        glVertex2f(
                      e12 + (radius * cos(i *  twicePi / triangleAmount)),
                          f12+ (radius * sin(i * twicePi / triangleAmount))
                        );
                }
        glEnd();
         glBegin(GL_QUADS);
glColor3ub(51, 204, 51);
glVertex2f(-0.85f,0.33f);
glVertex2f(-0.85f,0.44f);
glVertex2f(-0.65f,0.44f);
glVertex2f(-0.65f,0.33f);
glEnd();

GLfloat e123=-.8f; GLfloat f123=.5f;

        glBegin(GL_TRIANGLE_FAN);
          glColor3ub(51, 204, 51);
                glVertex2f(e123, f123); // center of circle
                for(i = 0; i <= triangleAmount;i++) {
                        glVertex2f(
```

```
                                    e123 + (radius * cos(i *  twicePi / triangleAmount)),
                                        f123+ (radius * sin(i * twicePi / triangleAmount))
                                      );
                              }

}
void hut()
{
   glBegin(GL_POLYGON);
   glColor3ub(255, 255, 0);
  // glColor3ub(204, 153, 0);
   glVertex2f(-0.5f,0.2f);
   glVertex2f(-0.1f,0.2f);
   glVertex2f(-0.18f,0.5f);
   glVertex2f(-0.58f,0.5f);
   glEnd();

    glBegin(GL_POLYGON);
    glColor3ub(255, 255, 0);
  // glColor3ub(204, 153, 0);
   glVertex2f(-0.49f,0.2f);
   glVertex2f(-0.13f,0.2f);
   glVertex2f(-0.13f,-0.2f);
   glVertex2f(-0.49f,-0.2f);
   glEnd();

    glBegin(GL_POLYGON);
    glColor3ub(192,201,200);
   glVertex2f(-0.58f,0.5f);
   glVertex2f(-0.63f,0.2f);
   glVertex2f(-0.61f,0.2f);
```

```
glVertex2f(-0.61f,-0.15f);
glVertex2f(-0.49f,-0.2f);
glVertex2f(-0.17f,-0.2f);

glEnd();
glLineWidth(2);
 glBegin(GL_LINES);
glColor3ub(0,0,0);
glVertex2f(-0.5f,0.2f);
glVertex2f(-0.1f,0.2f);

glColor3ub(0,0,0);
glVertex2f(-0.49f,0.2f);
glVertex2f(-0.49f,-0.2f);

glColor3ub(0,0,0);
 glVertex2f(-0.13f,-0.2f);
glVertex2f(-0.49f,-0.2f);

 glColor3ub(0,0,0);
 glVertex2f(-0.13f,-0.2f);
glVertex2f(-0.13f,0.2f);

glColor3ub(0,0,0);
glVertex2f(-0.5f,0.2f);
glVertex2f(-0.58f,0.5f);

glColor3ub(0,0,0);
glVertex2f(-0.58f,0.5f);
glVertex2f(-0.63f,0.2f);
```

```
glColor3ub(0,0,0);
glVertex2f(-0.57f,0.445f);
glVertex2f(-0.61f,0.2f);

glColor3ub(0,0,0);
glVertex2f(-0.61f,0.2f);
glVertex2f(-0.61f,-0.15f);

glColor3ub(0,0,0);
glVertex2f(-0.61f,-0.15f);
glVertex2f(-0.49f,-0.2f);

glColor3ub(0,0,0);
glVertex2f(-0.1f,0.2f);
glVertex2f(-0.18f,0.5f);

glColor3ub(0,0,0);
glVertex2f(-0.18f,0.5f);
glVertex2f(-0.58f,0.5f);

glEnd();

///DOOR1
 glBegin(GL_POLYGON);
 glColor3ub(255,255,0);

glVertex2f(-0.35f,0.1f);
glVertex2f(-0.22f,0.1f);
glVertex2f(-0.22f,-0.2f);
glVertex2f(-0.35f,-0.2f);
glEnd();
```

```
glLineWidth(2);
glBegin(GL_LINES);
glColor3ub(0,0,0);
glVertex2f(-0.35f,0.1f);
glVertex2f(-0.22f,0.1f);
glVertex2f(-0.22f,0.1f);
glVertex2f(-0.22f,-0.2f);
glVertex2f(-0.35f,0.1f);
glVertex2f(-0.35f,-0.2f);
glVertex2f(-0.285f,0.1f);
glVertex2f(-0.285f,-0.2f);
glEnd();

glBegin(GL_POLYGON);
glColor3ub(255, 255, 0);
glVertex2f(-0.44f,0.05f);
glVertex2f(-0.38f,0.05f);
glVertex2f(-0.38f,-0.05f);
glVertex2f(-0.44f,-0.05f);
glEnd();
 glLineWidth(2);
glBegin(GL_LINES);
glColor3ub(0,0,0);
glVertex2f(-0.44f,0.05f);
glVertex2f(-0.44f,-0.05f);
glVertex2f(-0.38f,0.05f);
glVertex2f(-0.38f,-0.05f);
glVertex2f(-0.44f,0.05f);
glVertex2f(-0.38f,0.05f);
glVertex2f(-0.44f,-0.05f);
glVertex2f(-0.38f,-0.05f);
```

```
    glVertex2f(-0.41f,0.05f);
    glVertex2f(-0.41f,-0.05f);
    glEnd();

glBegin(GL_POLYGON);
     glColor3ub(255,255,0);
     glVertex2f(-0.51f,0.12f);
    glVertex2f(-0.58f,0.14f);
    glVertex2f(-0.58f,-0.17f);
    glVertex2f(-0.51f,-0.2f);
    glEnd();
    glLineWidth(2);
    glBegin(GL_LINES);
    glColor3ub(0,0,0);
    glVertex2f(-0.58f,0.14f);
    glVertex2f(-0.58f,-0.17f);
    glVertex2f(-0.51f,0.12f);
    glVertex2f(-0.51f,-0.2f);
    glVertex2f(-0.58f,0.14f);
    glVertex2f(-0.51f,0.12f);
    glVertex2f(-0.545f,0.13f);
    glVertex2f(-0.545f,-0.185f);
    glEnd();

     glBegin(GL_POLYGON);
     glColor3ub(255, 255, 0);
    glVertex2f(-0.14f,0.05f);
    glVertex2f(-0.2f,0.05f);
    glVertex2f(-0.2f,-0.05f);
    glVertex2f(-0.14f,-0.05f);
    glEnd();
```

```
 glLineWidth(2);
glBegin(GL_LINES);
glColor3ub(0,0,0);
glVertex2f(-0.14f,0.05f);
glVertex2f(-0.14f,-0.05f);
glVertex2f(-0.2f,0.05f);
glVertex2f(-0.2f,-0.05f);
glVertex2f(-0.14f,0.05f);
glVertex2f(-0.2f,0.05f);
glVertex2f(-0.14f,-0.05f);
glVertex2f(-0.2f,-0.05f);
glVertex2f(-0.17f,0.05f);
glVertex2f(-0.17f,-0.05f);
glEnd();

}

void hut2()
{
glTranslatef(-.7,0,0);
glScaled(.7,.7,0);
glBegin(GL_POLYGON);
glColor3ub(192,201,200);
glVertex2f(-0.5f,0.2f);
glVertex2f(-0.1f,0.2f);
glVertex2f(-0.18f,0.5f);
glVertex2f(-0.58f,0.5f);
glEnd();

 glBegin(GL_POLYGON);
 glColor3ub(192,201,200);
```

```
glVertex2f(-0.49f,0.2f);
glVertex2f(-0.13f,0.2f);
glVertex2f(-0.13f,-0.2f);
glVertex2f(-0.49f,-0.2f);
glEnd();

 glBegin(GL_POLYGON);
 glColor3ub(192,201,200);
glVertex2f(-0.58f,0.5f);
glVertex2f(-0.63f,0.2f);
glVertex2f(-0.61f,0.2f);
glVertex2f(-0.61f,-0.15f);
glVertex2f(-0.49f,-0.2f);
glVertex2f(-0.17f,-0.2f);


glEnd();
glLineWidth(2);
 glBegin(GL_LINES);
glColor3ub(0,0,0);
glVertex2f(-0.5f,0.2f);
glVertex2f(-0.1f,0.2f);

glColor3ub(0,0,0);
glVertex2f(-0.49f,0.2f);
glVertex2f(-0.49f,-0.2f);

glColor3ub(0,0,0);
 glVertex2f(-0.13f,-0.2f);
glVertex2f(-0.49f,-0.2f);

 glColor3ub(0,0,0);
```

```
 glVertex2f(-0.13f,-0.2f);
glVertex2f(-0.13f,0.2f);

glColor3ub(0,0,0);
glVertex2f(-0.5f,0.2f);
glVertex2f(-0.58f,0.5f);

glColor3ub(0,0,0);
glVertex2f(-0.58f,0.5f);
glVertex2f(-0.63f,0.2f);

glColor3ub(0,0,0);
glVertex2f(-0.57f,0.445f);
glVertex2f(-0.61f,0.2f);

glColor3ub(0,0,0);
glVertex2f(-0.61f,0.2f);
glVertex2f(-0.61f,-0.15f);

glColor3ub(0,0,0);
glVertex2f(-0.61f,-0.15f);
glVertex2f(-0.49f,-0.2f);

glColor3ub(0,0,0);
glVertex2f(-0.1f,0.2f);
glVertex2f(-0.18f,0.5f);

glColor3ub(0,0,0);
glVertex2f(-0.18f,0.5f);
glVertex2f(-0.58f,0.5f);
```

```
glEnd();

 glBegin(GL_POLYGON);
 ///DOOR1
 glColor3ub(255,255,0);
glVertex2f(-0.35f,0.1f);
glVertex2f(-0.22f,0.1f);
glVertex2f(-0.22f,-0.2f);
glVertex2f(-0.35f,-0.2f);
glEnd();
glLineWidth(2);
glBegin(GL_LINES);
glColor3ub(0,0,0);
glVertex2f(-0.35f,0.1f);
glVertex2f(-0.22f,0.1f);
glVertex2f(-0.22f,0.1f);
glVertex2f(-0.22f,-0.2f);
glVertex2f(-0.35f,0.1f);
glVertex2f(-0.35f,-0.2f);
glVertex2f(-0.285f,0.1f);
glVertex2f(-0.285f,-0.2f);
glEnd();

glBegin(GL_POLYGON);
glColor3ub(255, 255, 0);
// glColor3ub(153, 115, 0);
glVertex2f(-0.44f,0.05f);
glVertex2f(-0.38f,0.05f); //left window
glVertex2f(-0.38f,-0.05f);
glVertex2f(-0.44f,-0.05f);
glEnd();
```

```
 glLineWidth(2);
glBegin(GL_LINES);
glColor3ub(0,0,0);
glVertex2f(-0.44f,0.05f);
glVertex2f(-0.44f,-0.05f);
glVertex2f(-0.38f,0.05f);
glVertex2f(-0.38f,-0.05f);
glVertex2f(-0.44f,0.05f);
glVertex2f(-0.38f,0.05f);
glVertex2f(-0.44f,-0.05f);
glVertex2f(-0.38f,-0.05f);
glVertex2f(-0.41f,0.05f);
glVertex2f(-0.41f,-0.05f);
glEnd();

glBegin(GL_POLYGON);
  glColor3ub(255, 255, 0);
//glColor3ub(153, 115, 0);
 glVertex2f(-0.51f,0.12f); //2nd door
glVertex2f(-0.58f,0.14f);
glVertex2f(-0.58f,-0.17f);
glVertex2f(-0.51f,-0.2f);
glEnd();
glLineWidth(2);
glBegin(GL_LINES);
glColor3ub(0,0,0);
glVertex2f(-0.58f,0.14f);
glVertex2f(-0.58f,-0.17f);
glVertex2f(-0.51f,0.12f);
glVertex2f(-0.51f,-0.2f);
glVertex2f(-0.58f,0.14f);
```

```
    glVertex2f(-0.51f,0.12f);
    glVertex2f(-0.545f,0.13f);
    glVertex2f(-0.545f,-0.185f);
    glEnd();


     glBegin(GL_POLYGON);
     glColor3ub(255, 255, 0);
    //glColor3ub(153, 115, 0);
    glVertex2f(-0.14f,0.05f);
    glVertex2f(-0.2f,0.05f);   //rightwindow
    glVertex2f(-0.2f,-0.05f);
    glVertex2f(-0.14f,-0.05f);
    glEnd();
     glLineWidth(2);
    glBegin(GL_LINES);
    glColor3ub(0,0,0);
    glVertex2f(-0.14f,0.05f);
    glVertex2f(-0.14f,-0.05f);
    glVertex2f(-0.2f,0.05f);
    glVertex2f(-0.2f,-0.05f);
    glVertex2f(-0.14f,0.05f);
    glVertex2f(-0.2f,0.05f);
    glVertex2f(-0.14f,-0.05f);
    glVertex2f(-0.2f,-0.05f);
    glVertex2f(-0.17f,0.05f);
    glVertex2f(-0.17f,-0.05f);
    glEnd();
    glLoadIdentity();


}
```

```
void boat()
{
glTranslatef(moveb,-0.1f,0.0f);
   glBegin(GL_POLYGON);
            glColor3ub(0,0,0);
            glVertex2f(-0.2f, 0.4f);
            glVertex2f(-0.15f, 0.35f);
            glVertex2f(0.15f, 0.35f);
   glVertex2f(0.2f, 0.4f);
   glEnd();

            glBegin(GL_POLYGON);
            glColor3ub(255, 153, 0);
            glVertex2f(-0.13f, 0.4f);
            glVertex2f(-0.11f,0.48f);
            glVertex2f(-0.088f, 0.52f);
            glVertex2f(0.13f, 0.52f);
            glVertex2f(0.14f, 0.49f);
   glVertex2f(0.15f, 0.4f);
   glEnd();

            glBegin(GL_POLYGON);
            glColor3ub(255,25,25);
            glVertex2f(-0.038f, 0.57f);
            glVertex2f(-0.038f, 0.73f);
            glVertex2f(-0.035f, 0.75f);
            glVertex2f(0.064f, 0.73f);
            glVertex2f(0.065f, 0.71f);
   glVertex2f(0.065f, 0.55f);
```

```
    glEnd();

    glBegin(GL_POLYGON);
            glColor3ub(136,204,0);
            glVertex2f(0.0f, 0.52f);
            glVertex2f(0.0f, 0.79f);
            glVertex2f(0.01f, 0.79f);
            glVertex2f(0.01f, 0.52f);
            glEnd();
}

void boat3()
{


    glBegin(GL_POLYGON);
            glColor3ub(0,0,0);
            glVertex2f(0.4f, -0.15f);
            glVertex2f(0.45f, -0.2f);
            glVertex2f(0.75f, -0.2f);
    glVertex2f(0.8f, -0.15f);
    glEnd();

    glBegin(GL_POLYGON);
            glColor3ub(239, 108, 0);
            glVertex2f(0.47f, -0.15f);
            glVertex2f(0.49f,-0.15f);
            glVertex2f(0.512f, -0.12f);
            glVertex2f(0.73f, -0.12f);
            glVertex2f(0.74f, -0.15f);
    glVertex2f(0.75f, -0.15f);
```

```
   glEnd();



            glBegin(GL_POLYGON);
            glColor3ub(233, 30, 99 );
            glVertex2f(0.462f, -0.08f);
            glVertex2f(0.462f, 0.08f);
            glVertex2f(0.465f, 0.1f);
            glVertex2f(0.564f, 0.08f);
            glVertex2f(0.565f, 0.06f);
   glVertex2f(0.565f, -0.1f);
   glEnd();

   glBegin(GL_POLYGON);
            glColor3ub(123, 36, 28);
            glVertex2f(0.5f, -0.13f);
            glVertex2f(0.5f, 0.14f);
            glVertex2f(0.51f, 0.14f);
            glVertex2f(0.51f, -0.13f);
            glEnd();
}






void way()
{
   glBegin(GL_QUADS);
```

```
    glColor3ub(153, 153, 102);
    glVertex2f(-0.35f,-0.2f);
    glVertex2f(-0.22f,-0.2f);
    glVertex2f(-0.28f,-0.5f);
    glVertex2f(-0.43f,-0.5f);
    glEnd();

    glBegin(GL_QUADS);
    glColor3ub(153, 153, 102);
    glVertex2f(-0.43f,-0.5f);
    glVertex2f(-0.75f,-1.0f);
    glVertex2f(-0.56f,-1.0f);
    glVertex2f(-0.28f,-0.5f);
    glEnd();



}
void way2()
{
    glTranslatef(-.7,-0.00,0);
    glScaled(.7,.7,0);
    glBegin(GL_QUADS);
    glColor3ub(153, 153, 102);
    glVertex2f(-0.35f,-0.2f);
    glVertex2f(-0.22f,-0.2f);
    glVertex2f(-0.28f,-0.5f);
    glVertex2f(-0.43f,-0.5f);
    glEnd();

    glBegin(GL_QUADS);
    glColor3ub(153, 153, 102);//153, 153, 102
```

```
    glVertex2f(-0.43f,-0.5f);
    glVertex2f(0.45f,-.7f);
    glVertex2f(0.36f,-.8f);
    glVertex2f(-0.28f,-0.5f);
    glEnd();
    glLoadIdentity();



}
void fence()
{
     glLineWidth(4);
     glBegin(GL_LINES);
    glColor3ub(255, 255, 102);
    glVertex2f(-1.0f,-0.1f);
    glVertex2f(-0.6f,-0.1f);

    glColor3ub(255, 255, 102);
    glVertex2f(-1.0f,-0.05f);
    glVertex2f(-0.6f,-0.05f);

    glColor3ub(255, 255, 102);
    glVertex2f(-1.0f,0.0f);
    glVertex2f(-0.6f,0.0f);

     glColor3ub(255, 255, 102);
    glVertex2f(-1.0f,0.05f);
    glVertex2f(-0.6f,0.05f);

     glColor3ub(255, 255, 102);
    glVertex2f(-1.0f,0.1f);
```

```
glVertex2f(-0.6f,0.1f);

 glColor3ub(255, 255, 102);
glVertex2f(-0.95f,0.13f);
glVertex2f(-0.95f,-0.12f);

glColor3ub(255, 255, 102);
glVertex2f(-0.9f,0.13f);
glVertex2f(-0.9f,-0.12f);

glColor3ub(255, 255, 102);
glVertex2f(-0.85f,0.13f);
glVertex2f(-0.85f,-0.12f);

glColor3ub(255, 255, 102);
glVertex2f(-0.8f,0.13f);
glVertex2f(-0.8f,-0.12f);

glColor3ub(255, 255, 102);
glVertex2f(-0.75f,0.13f);
glVertex2f(-0.75f,-0.12f);

glColor3ub(255, 255, 102);
glVertex2f(-0.7f,0.13f);
glVertex2f(-0.7f,-0.12f);

glColor3ub(255, 255, 102);
glVertex2f(-0.65f,0.13f);
glVertex2f(-0.65f,-0.12f);

glEnd();
```

```
}

void cow()
{

    glTranslatef(-4.88,-2.5,0);
    glScaled(4.8,4.8,0);

      float theta;
     //up
     glColor3ub(40,40,40);
    glBegin(GL_POLYGON);
    for( int i=0;i<360;i++)
    {
       theta=i*3.142/180;
       glVertex2f(0.899+0.007*cos(theta),0.487+0.009*sin(theta));
    }
    glEnd();
   //cow 1
    glColor3ub(40,40,40);
    glBegin(GL_POLYGON);
    glVertex3f(0.89,0.47,0.0);
    glVertex3f(0.93,0.47,0.0);
    glVertex3f(0.93,0.49,0.0);
    glVertex3f(0.89,0.49,0.0);
    glEnd();

    //Head
    glColor3ub(255,255,255);
    glBegin(GL_POLYGON);
```

```
glVertex3f(0.882,0.47,0.0);
glVertex3f(0.885,0.47,0.0);
glVertex3f(0.887,0.49,0.0);
glVertex3f(0.878,0.49,0.0);
glEnd();

//throat
glColor3ub(40,40,40);
glBegin(GL_POLYGON);
glVertex3f(0.88,0.483,0.0);
glVertex3f(0.89,0.483,0.0);
glVertex3f(0.89,0.49,0.0);
glVertex3f(0.88,0.49,0.0);
glEnd();

//Tail
glColor3ub(255,255,255);
glBegin(GL_LINES);
glVertex3f(0.937,0.47,0.0);
glVertex3f(0.928,0.49,0.0);
glEnd();



glColor3ub(40,40,40);
glBegin(GL_POLYGON);
glVertex3f(0.935,0.467,0.0);
glVertex3f(0.938,0.467,0.0);
glVertex3f(0.938,0.472,0.0);
glVertex3f(0.935,0.472,0.0);
glEnd();
```

```
    //Legs
    glColor3ub(40,40,40);
    glBegin(GL_LINES);
    glVertex3f(0.927,0.45,0.0);
    glVertex3f(0.929,0.47,0.0);
    glEnd();

    glColor3ub(40,40,40);
    glBegin(GL_LINES);
    glVertex3f(0.924,0.45,0.0);
    glVertex3f(0.927,0.47,0.0);
    glEnd();

    glColor3ub(40,40,40);
    glBegin(GL_LINES);
    glVertex3f(0.889,0.45,0.0);
    glVertex3f(0.89,0.47,0.0);
    glEnd();

    glColor3ub(40,40,40);
    glBegin(GL_LINES);
    glVertex3f(0.897,0.45,0.0);
    glVertex3f(0.89,0.47,0.0);
    glEnd();
      glLoadIdentity();

}

void StartingText()
{
```

```
    char text[120];
    sprintf(text, "VILLAGE SCENERY",5.00,8.00);
    glColor3f(0, 0, 1);
    glRasterPos2f( -20 , 12 );
    for(int i = 0; text[i] != '\0'; i++)
    {
        if(text[i]==' '  && text[i+1]==' ')
        {
            glutBitmapCharacter(GLUT_BITMAP_TIMES_ROMAN_24, text[i]);
            glRasterPos2f( -32 , 02 );
        }
        else glutBitmapCharacter(GLUT_BITMAP_TIMES_ROMAN_24, text[i]);
    }
    sprintf(text,"PRESS S FOR START",5.00,8.00);
    glColor3f(0, 0, 0);
    glRasterPos2f( -22 , 3 );
    for(int i = 0; text[i] != '\0'; i++)
    {
         if(text[i]==' '  && text[i+1]==' ')
        {
            glutBitmapCharacter(GLUT_BITMAP_TIMES_ROMAN_24, text[i]);
            glRasterPos2f( -31 , 02 );
        }
        else glutBitmapCharacter(GLUT_BITMAP_TIMES_ROMAN_24, text[i]);
    }

}

void updateb(int value) {
moveb += 0.00;
if(moveb > 1)
```

```
{
moveb = 1;
}

else if(moveb < -1)
{
moveb= -1;
}
moveb += speedb;
glutPostRedisplay();

 glutTimerFunc(20, updateb, 0);
}



void display() {
        glClearColor(0.0f, 0.0f, 0.0f, 1.0f);
        glClear(GL_COLOR_BUFFER_BIT);
        glMatrixMode(GL_PROJECTION);
        glLoadIdentity();
        glLineWidth(2);
   sky();
   sun();
   glPushMatrix();
   glTranslatef(position2,0.0f, 0.0f);
   cloud1();
   cloud2();
   glPopMatrix();
   glPushMatrix();
   glTranslatef(position22,0.0f, 0.0f);
   bird();
```

```
    glPopMatrix();

    hills();

    river();

    glPushMatrix();


    boat();

    glPopMatrix();

    ground();

    way();

    way2();


    boat3();

    fence();

    hut2();

    tree();

    hut();

    cow();

if(sunPosition < -.57)

            {

                moon();

            }

else if(sunPosition < -.55)

{

      h=0.0;

                        i=0.0;

                        j=0.0;


}


            glFlush();

            glutSwapBuffers();
```

```
}

void reshape(int w, int h)
{

   float aspectRatio = (float)w/(float)h;
      glMatrixMode(GL_PROJECTION);
   glLoadIdentity();
      gluPerspective(145, aspectRatio, 1.0, 100.0);
   glMatrixMode(GL_MODELVIEW);

}

void Display(void)
{

   glClear(GL_COLOR_BUFFER_BIT | GL_DEPTH_BUFFER_BIT);
   glLoadIdentity();
   glTranslatef(0,0,-20);
   StartingText();
   glFlush();
   glutSwapBuffers();
}

void init(void)
{
   glClearColor( 1.0f, 1.0f, 1.0f, 1.0f);
   glClearDepth( 1.0 );
   glEnable(GL_DEPTH_TEST);
   glEnable(GL_LIGHTING);
   glShadeModel(GL_SMOOTH);
```

```
    glEnable(GL_COLOR_MATERIAL);
    glColorMaterial(GL_FRONT, GL_AMBIENT_AND_DIFFUSE);
    glEnable(GL_LIGHT0);



}
void handleKeypress(unsigned char key, int x, int y) {
            switch (key) {
               case  's':
                  glutDestroyWindow(1);
                  glutInitWindowSize(1240, 750);
        glutInitWindowPosition((glutGet(GLUT_SCREEN_WIDTH)-1240)/2,(glutGet(GLUT_SCREEN_HEIGHT)-750)/2);
        glutCreateWindow("village scenery");
        glutKeyboardFunc(handleKeypress);
        glutDisplayFunc(display);
        break;



     case 'l':


    speedb = -0.002;
    break;
    case 'r':


    speedb = 0.002;
    break;

glutPostRedisplay();


    }
}
```

```
int main(int argc, char** argv)
{
  glutInit(&argc, argv);
  glutInitWindowSize(1240, 750);
  glutInitWindowPosition((glutGet(GLUT_SCREEN_WIDTH)-1240)/2,(glutGet(GLUT_SCREEN_HEIGHT)-750)/2);
  glutCreateWindow("village scenery");
  init();
  glutReshapeFunc(reshape);
  glutDisplayFunc(Display);
  glutTimerFunc(100, cloud, 0);
  glutTimerFunc(100, birdd, 0);
  glutTimerFunc(1, updateb, 0);
  glutTimerFunc(1000, updateSun, 0);
  glutTimerFunc(1000, updateMoon, 0);
  glutKeyboardFunc(handleKeypress);
  init();
  glutMainLoop();
  return 0;
}
```

Project benefit:

This is a scenario of a typical village where people can live and get clear and fresh weather. The rural climate is very important to get rid of the polluted urban environment. The fields are used as children’s playground in rural area that we don’t usually find in urban life. People can get pollution free milk from domesticated cows. They also can be able to eat fishes from river. Iron-free fresh river water can be used in household works. People will able to travel by boat in the rainy season. We have tried to create a healthy environment through this project.
